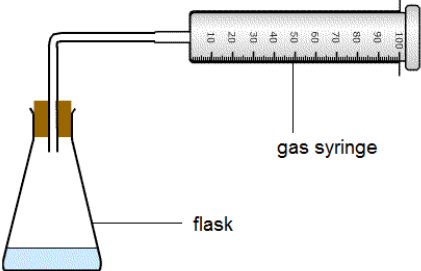


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			 flask with stopper and delivery tube (1) method of collecting and measuring volume of oxygen (1) accept collection over water using measuring cylinder	2			2		2
	(b)	(i)		tangent drawn at $t = 0$ or $t = 200$ s (1) initial rate = $10\text{--}15 \text{ cm}^3\text{s}^{-1}$ (1)		2		2	2	2
		(ii)		rate at $t = 200\text{s} = 3\text{--}5 \text{ cm}^3\text{s}^{-1}$		1		1	1	1
		(iii)		at $t = 200$ concentration lower / particles further apart (1) less frequent collisions / less chance of collisions, so rate less (1)	2			2		
		(iv)		2000 cm^3 oxygen = $2000/24500 \text{ mol} = 0.0816$ (1) number of moles $\text{H}_2\text{O}_2 = 0.163$ (1) concentration = 1.63 – must be correct to 3 sig figs (1)		2	1	3	1	1